

- 7 Fig. 7.1 shows part of a section through the leaf of a C₄ plant such as maize. The letters **A**, **B** and **C** show three types of cell found in the leaf.

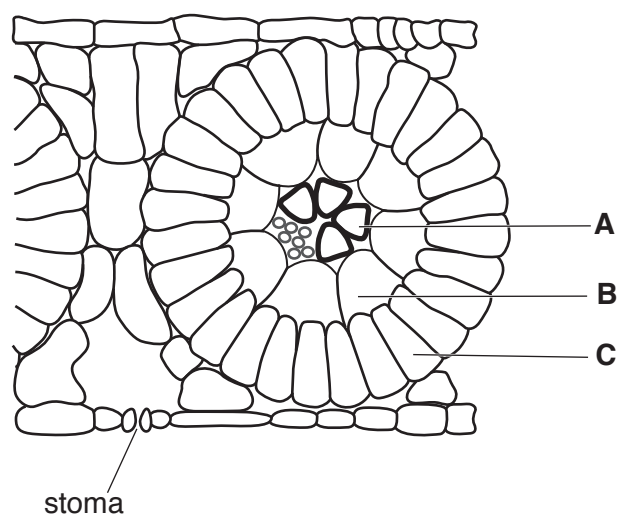


Fig. 7.1

- (a) Complete Table 7.1 by using the letters **A**, **B** or **C** from Fig. 7.1 to show the location of several compounds associated with photosynthesis in C₄ plants.

You may use **A**, **B** and **C** once, more than once, or not at all.

Table 7.1

compound	location
PEP carboxylase	
RuBP	
rubisco	

[3]

- (b) Explain why the cells in **C** form a tight ring around the cells in **B**.

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[2]

- (c)** Several enzymes have a role in the light independent stage of photosynthesis.

Fig. 7.2 shows the activity of one of these enzymes at different temperatures for a C3 plant, *Pisum sativum*, and a C4 plant, *Amaranthus hypochondriacus*.

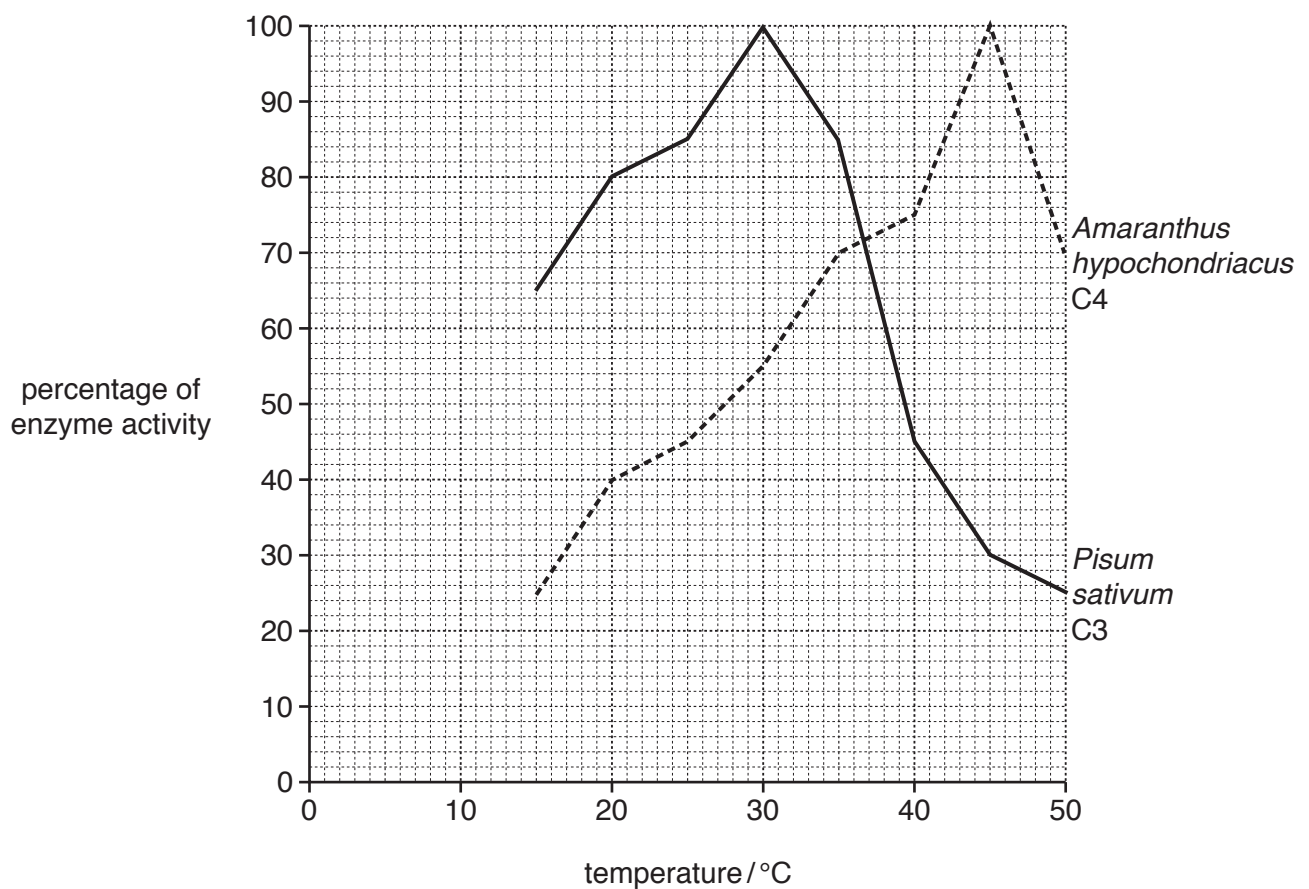


Fig. 7.2

Describe the differences between the two curves **and** suggest explanations for these differences.

.....[3]